

Basics of FE-Analysis

BK10A6400

Lecture 4: Symmetric boundary conditions, beam element

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Today's agenda

- Symmetric boundary conditions
- Beam-like structures
- Simple beam element
- Displacement field - strain - stress - energy - stiffness matrix
- Example: cantilever problem
- Bonus Task 4: a simple beam structure.



Robert D. Cook, David S. Malkus, Michale E. Plesha,
Robert J. Watt

Concepts and applications of finite element analysis
4th edition, *Wiley (2002)*.



Matti K. Hakala

Lujuusopin elementtimenetelmä
Otapaino (1980).

Rod/truss element

- Only axial displacement field approximation.
- One DOF at node.
- Linear interpolation $u_h = N_1 u_1 + N_2 u_2 = \mathbf{N} \mathbf{u}$.

$$\mathbf{N} = \begin{pmatrix} N_1 & N_2 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad (1)$$

$$N_1 = 1 - x/L \quad ; \quad N_2 = x/L \quad (2)$$

$$\overline{\mathbf{K}} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad (3)$$

Properties of elemental stiffness matrices: Examples

The elemental stiffness matrix for a rod/truss element:

$$\overline{\mathbf{K}} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad (4)$$

- Symmetric
- Determinant: $\det \overline{\mathbf{K}} = 0$
- A stiffness matrix of a rod element has one rigid body mode (in the axial direction).
- Matrix rank: $\text{rank } \overline{\mathbf{K}} = 1$

Simple examples: how to use MATLAB symbolic toolbox?

- MATLAB codes are shared with the lecture slides in Moodle.
- Remember not to mix the symbolic and numeric parts of the codes in the same m-file.
- Symbolic syntax differs often from numeric ones.

Static

- "Practical drawing" convention: the bending moment is positive if the beam's bottom surface is in tension ("smile").
- → Easy to visualize, suitable for hand calculations and diagrams.

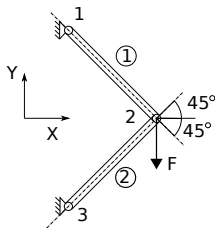
FEM

- Mathematical convention: the right-hand rule defines the sign of forces and moments.
- → Easy to implement in programming and to assemble matrices.

Sign conventions: to be completed in the lecture

Recap: Example: Solve displacements at node 2 of a simple truss structure

Solve displacements with rod finite elements.

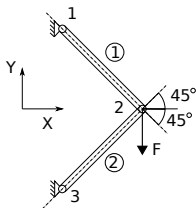


- Global nodal displacements (u_X, u_Y) at each node.
- 3 nodes, 2 DOFs at node $\rightarrow$ a system of 6 DOF before taking into account the (linear) constraints that are applied by eliminating rows and columns of the stiffness matrix for a system.

- Constraints i.e. fixed DOF's:

$$u_{X_1} = 0, u_{Y_1} = 0, u_{X_3} = 0, u_{Y_3} = 0$$

Example: Solve displacements at node 2 of a simple truss structure



The parameters of the problem are given in Table.

	Young's modulus, E [MPa]	Cross section, A [mm ²]	Length, L [mm]
Truss 1	210000	100	1000
Truss 2	210000	50	1000
Force, F [N]	100 000		

Example: Simple truss structure problem

- Elem 1 consists of nodes 1 and 2
- Elem 2 consists of nodes 2 and 3
- Angle of a rod 1: ?
- Angle of a rod 2: ?

Transformation into a global coordinates system.

Rod 1:

$$\mathbf{K}_G^{(1)} = \mathbf{T}(\theta_1)^T \overline{\mathbf{K}} \mathbf{T}(\theta_1) \quad (5)$$

Rod 2:

$$\mathbf{K}_G^{(2)} = \mathbf{T}(\theta_2)^T \overline{\mathbf{K}} \mathbf{T}(\theta_2) \quad (6)$$

Example: Simple truss structure problem: Connectivity

Sometimes it is helpful to present elements' connectivity and corresponding nodal displacements in the table as follows:

	Node I	Node J	nodal displacements
Truss 1	1	2	u_1, v_1, u_2, v_2
Truss 2	2	3	u_2, v_2, u_3, v_3

It can be seen from the connectivity table what stiffness components of global elemental stiffness matrices $\mathbf{K}^{(1)}$ and $\mathbf{K}^{(2)}$ are needed to sum to mimic the structure correctly. After assembling, the global stiffness matrix should be a size of 6×6 .

Rod 1:

$$\mathbf{K}_G^{(1)} = \mathbf{T}(\theta_1)^T \overline{\mathbf{K}} \mathbf{T}(\theta_1) \quad (7)$$

$$\mathbf{K}_G^{(1)} = \begin{matrix} & \begin{matrix} u_1 & v_1 & u_2 & v_2 \end{matrix} \\ \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} \end{pmatrix} & \begin{matrix} u_1 \\ v_2 \\ u_2 \\ v_2 \end{matrix} \end{matrix}$$

Rod 2:

$$\mathbf{K}_G^{(2)} = \mathbf{T}(\theta_2)^T \overline{\mathbf{K}} \mathbf{T}(\theta_2) \quad (8)$$

$$\mathbf{K}_G^{(2)} = \begin{matrix} & \begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{pmatrix} k_{11}^{(2)} & k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} \\ k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} \\ k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} \end{pmatrix} & \begin{matrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix} \end{matrix}$$

Assembling

$$\mathbf{K}_G = \begin{matrix} & \begin{matrix} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} & 0 & 0 \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} & 0 & 0 \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} + k_{11}^{(2)} & k_{34}^{(1)} + k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} + k_{21}^{(2)} & k_{44}^{(1)} + k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} \\ 0 & 0 & k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} \\ 0 & 0 & k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} \end{pmatrix} & \begin{matrix} u_1 \\ v_1 \\ u_2 \\ u_2 \\ u_3 \\ v_3 \end{matrix} \end{matrix}$$

External forces are set up in a global coordinate system because our global stiffness matrix is also presented in a global coordinate system.

Force F is active at node 2 in the same direction with nodal displacement v_2 .

$$\mathbf{f}_G = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -F \\ 0 \\ 0 \end{pmatrix} \begin{matrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}$$

Boundary conditions

Fixed nodal displacements are $u_1 = v_1 = u_3 = v_3 = 0$. Let's eliminate corresponding rows and columns regarding FE-matrices.

$$\mathbf{f}_{G_{red}} = \begin{pmatrix} 0 \\ -F \end{pmatrix} \begin{matrix} u_2 \\ v_2 \end{matrix}$$

$$\mathbf{K}_{G_{red}} = \begin{matrix} & \begin{matrix} u_2 & v_2 \end{matrix} \\ \begin{pmatrix} k_{33}^{(1)} + k_{11}^{(2)} & k_{34}^{(1)} + k_{12}^{(2)} \\ k_{43}^{(1)} + k_{21}^{(2)} & k_{44}^{(1)} + k_{22}^{(2)} \end{pmatrix} & \begin{matrix} u_2 \\ v_2 \end{matrix} \end{matrix}$$

Solving global displacements

Note that $\mathbf{K}_{G_{red}}$ is symmetric i.e. $k_{ij} = k_{ji}$ (if not, there is a typo) and $\det \mathbf{K}_{G_{red}} > 0$ (if not, check BC's).

And then displacements (in a global coordinate system) can be solved as

$$\mathbf{u}_{G_{red}} = \mathbf{K}_{G_{red}}^{-1} \mathbf{f}_{G_{red}}$$

Elemental displacements and member forces

After solving $\mathbf{u}_{G_{red}}$, local elemental displacements (for an element i) can be computed as follows:

$$\bar{\mathbf{u}}^{(i)} = \mathbf{T}^{(i)} \mathbf{u}_G^{(i)} \quad (9)$$

Member forces (for an element i) can be found as

$$\bar{\mathbf{f}}^{(i)} = \bar{\mathbf{K}}^{(i)} \bar{\mathbf{u}}^{(i)} \quad (10)$$

Note $\mathbf{u}_G^{(i)}$ includes element's i (solved nodal coordinates and nodal coordinates at supports).

- Structures can be symmetric (or antisymmetric) -> could be used to reduce computational effort.
- The geometry, loading, and boundary conditions can be mirrored through a symmetry plane (or line).
- Less CPU time, but be sure your reduced structure describes the original structure.
- You need to consider the deflection curve to determine the equivalent reduced structure.

Example: a symmetric truss structure (two struts, ss-ss): half empty here, completed together in the lecture

Maxwell's stability:

m = number of **members** (bars) = 2

j = number of **joints** (nodes) = 3

r = number of **support reactions** = 4

Criterion:

$$m + r \begin{cases} < 2j & \text{mechanism (unstable)} \\ = 2j & \text{isostatic (stable)} \\ > 2j & \text{redundant} \end{cases}$$

For our simple structure: $m + r = 2j$

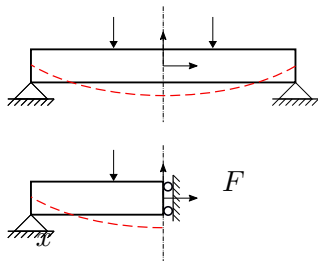
Example: a symmetric truss structure (two struts, ss-ss): half empty here, completed together in the lecture

- If a nodal force is located in the plane of sym. line, the force has to be divided by two.
- If a truss element is located in the plane (line) of sym., the cross-sectional area has to be divided by two.
- Now a simple truss structure is symmetric with respect to the yz -plane.

Sym. plane	u	v	w	θ_x	θ_y	θ_z
xy	Free	Free	Fix	Fix	Fix	Free
xz	Free	Fix	Free	Fix	Free	Fix
yz	Fix	Free	Free	Free	Fix	Fix

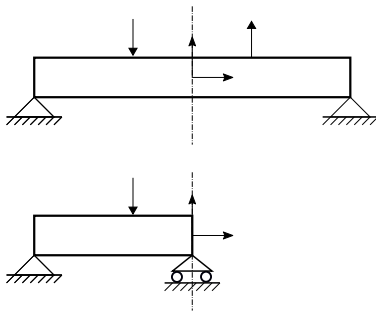
Example: a symmetric truss structure (two struts, ss-ss): half empty here, completed together in the lecture

Example: a symmetric beam structure (ss-ss)



- Vertical deflection free, rotation fixed.
- $u_x = 0, \theta_z = 0$ at symmetric line

Example: an antisymmetric beam structure (ss-ss)



- Vertical deflection fixed, rotation free.
- $u_y = 0$

Boundary conditions for symmetric loading

Sym. plane	u	v	w	θ_x	θ_y	θ_z
xy	Free	Free	Fix	Fix	Fix	Free
xz	Free	Fix	Free	Fix	Free	Fix
yz	Fix	Free	Free	Free	Fix	Fix

Boundary conditions for antisymmetric loading

Sym. plane	u	v	w	θ_x	θ_y	θ_z
xy	Fix	Fix	Free	Free	Free	Fix
xz	Fix	Free	Fix	Free	Fix	Free
yz	Free	Fix	Fix	Fix	Free	Free

Examples of beam structures

- Numerous applications in the construction and engineering industries.
- Shafts and axles
- Cranes
- Railway's rails (6 meter length per rail) and sleepers
- Wind turbine's blades

Some beam-type structures on a way to work



When can a structure be approximated as a beam?

- Every object is three-dimensional but it is still computationally too expensive in some large systems.
- Mathematical simplification
- If one dimension (length) is much larger than the other two (height and width of cross-section) then the beam element may be suitable.
- The user decides so the user should know the limitations.

RECAP from previous courses: Euler-Bernoulli beam theory

- Simplification to compute displacement of beams
- Small displacements (linear theory). Uses approximations for angles.
- Cross sections remain plane (straight)
- Cross section is perpendicular to the beam's neutral axis (i.e. no shear deformation).

RECAP: Euler–Bernoulli beam equation

For a beam with constant bending stiffness EI :

$$EI \frac{d^4 v(x)}{dx^4} = q(x) \quad (11)$$

The internal forces are obtained from the deflection:

$$M(x) = EI \frac{d^2 v}{dx^2}, \quad (12)$$

$$Q(x) = \frac{dM}{dx} = EI \frac{d^3 v}{dx^3}. \quad (13)$$

The deflection $v(x)$ is obtained by integrating the beam equation four times. The four integration constants are determined from the boundary conditions.

For example, for a cantilever beam:

$$v(0) = 0, \quad v'(0) = 0, \quad M(L) = 0, \quad Q(L) = 0. \quad (14)$$

RECAP: Euler–Bernoulli beam equation

For a beam with constant bending stiffness EI :

$$EI \frac{d^4 v(x)}{dx^4} = q(x) \quad (15)$$

The internal forces are

$$M(x) = EI \frac{d^2 v}{dx^2}, \quad Q(x) = EI \frac{d^3 v}{dx^3}. \quad (16)$$

For a cantilever beam subjected to a constant load q_0 :

$$v(0) = v'(0) = 0, \quad M(L) = Q(L) = 0. \quad (17)$$

The displacement solution is

$$v(x) = \frac{q_0 x^2}{24EI} (x^2 - 4Lx + 6L^2) \quad (18)$$

Euler–Bernoulli beam: point load at the free end

For a cantilever beam with a point load P at the free end,

$$EI \frac{d^4 v}{dx^4} = 0 \quad (19)$$

because there is no distributed load along the beam.

The boundary conditions are

$$v(0) = 0, \quad v'(0) = 0, \quad (20)$$

$$EI v''(L) = 0, \quad EI v'''(L) = P. \quad (21)$$

The analytical displacement solution is

$$v(x) = \frac{Px^2}{6EI} (3L - x) \quad (22)$$

The exact displacement is a cubic polynomial in x .

A two-node Euler–Bernoulli beam element also uses a cubic interpolation for $v(x)$. this displacement field exactly.

Two-noded 4 DOF Euler-Bernoulli beam element: interpolation

Two node 4 DOF Euler-Bernoulli type beam element has nodal displacement v_1 and rotation θ_1 at node 1 ($x = 0$) and v_2 and rotation θ_2 at node 2 ($x = L$). It is assumed that the element's local coordinate system $\{x, y\}$ is at an element's node 1 ($x = 0$). Four polynomials are needed to approximate displacement field v_h in the vertical direction over the element's length (material coordinate x).

$$v_h = a_0 + a_1x + a_2x^2 + a_3x^3 = \mathbf{a}^T \mathbf{p} \quad (23)$$

where $\mathbf{a} = [a_0 \ a_1 \ a_2 \ a_3]^T$ and $\mathbf{p} = [1 \ x \ x^2 \ x^3]^T$. Rotation θ can be approximated from the displacement field as follows:

$$\theta \approx \frac{dv_h}{dx} = a_1 + 2a_2x + 3a_3x^2 \quad (24)$$

Two-noded 4 DOF Euler-Bernoulli beam element: interpolation

Taking boundary conditions into account at elemental level

$v_h(x=0) := v_1$, $\theta(x=0) := \theta_1$, $v_h(x=L) := v_2$ and $\theta(x=L) := \theta_2$:

$$\begin{cases} v_h(0) := a_0 = v_1 \\ \theta(0) := a_1 = \theta_1 \\ v_h(L) := a_0 + a_1L + a_2L^2 + a_3L^3 = v_2 \\ \theta(L) := a_1 + 2a_2L + 3a_3L^2 = \theta_2 \end{cases} \quad (25)$$

Eqs. (25) in a matrix form:

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & L & L^2 & L^3 \\ 0 & 1 & 2L & 3L^2 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}}_{\mathbf{a}} = \underbrace{\begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix}}_{\mathbf{u}} \quad (26)$$

Two-noded 4 DOF Euler-Bernoulli beam element: shape functions

Solving $\mathbf{a} = \mathbf{A}^{-1}\mathbf{u}$ and substituting them into Eq.(23).

$$v_h = (\mathbf{A}^{-1}\mathbf{u})^T \mathbf{p} = \underbrace{\mathbf{p}^T \mathbf{A}^{-1}}_{\mathbf{N}^b} \mathbf{u} \quad (27)$$

Solving $\mathbf{N}^b = \mathbf{p}^T \mathbf{A}^{-1}$, shape functions can be written explicitly as

$$N_1 = \frac{2x^3}{L^3} - \frac{3x^2}{L^2} + 1 \quad ; N_2 = \frac{x^3}{L^2} - \frac{2x^2}{L} + x \quad ; N_3 = -\frac{2x^3}{L^3} + \frac{3x^2}{L^2} \quad ; N_4 = \frac{x^3}{L^2} - \frac{x^2}{L}$$

The displacement field is rewritten now:

where
$$v_h = \mathbf{N}^b \mathbf{u} \quad (28)$$

$$\mathbf{N}^b = \begin{pmatrix} N_1 & N_2 & N_3 & N_4 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} v_1 & \theta_1 & v_2 & \theta_2 \end{pmatrix}^T$$

4 DOF's Euler-Bernoulli beam element: Energy

Strain energy due to bending deformation:

$$W_{int}^{bend} = \frac{1}{2} \int_V \sigma \varepsilon \, dV = \frac{1}{2} E \int_0^L \varepsilon_{xx}^2 \int_A y^2 \, dA \, dx = \frac{1}{2} EI \int_0^L \varepsilon_{xx}^2 \, dx \quad (29)$$

where I is moment of inertia and ε_{xx}^1 is curvature.

$$\varepsilon_{xx}^1 \approx \frac{d\theta}{dx} = \frac{d^2 v_h}{dx^2} = \frac{\partial^2 \mathbf{N}^b}{\partial x^2} \mathbf{u}_b \quad (30)$$

$$W_{int}^{bend} = \frac{1}{2} \bar{\mathbf{u}}_b^T \underbrace{EI \int_0^L \frac{\partial^2 \mathbf{N}^{b^T}}{\partial x^2} \frac{\partial^2 \mathbf{N}^b}{\partial x^2} dx}_{\bar{\mathbf{K}}_{EB4DOF}} \bar{\mathbf{u}}_b \quad (31)$$

4 DOF's Euler-Bernoulli beam element: Derivation of stiffness matrix

so the stiffness matrix can be found directly from Eq. (31):

$$\overline{\mathbf{K}}_{EB4DOF} = EI \int_0^L \frac{\partial^2 \mathbf{N}^{bT}}{\partial x^2} \frac{\partial^2 \mathbf{N}^b}{\partial x^2} dx \quad (32)$$

or using a more general approach by defining the vector of elemental internal forces:

$$\overline{\mathbf{f}}_{int} = \frac{\partial W_{int}^{bend}}{\partial \overline{\mathbf{u}}} \quad (33)$$

and elemental local stiffness matrix:

$$\overline{\mathbf{K}}_{EB4DOF} = \frac{\partial \overline{\mathbf{f}}_{int}}{\partial \overline{\mathbf{u}}} \quad (34)$$

4 DOF Euler-Bernoulli beam element: Stiffness matrix

The elemental stiffness matrix for a 2D Euler-Bernoulli 4 DOF beam element:

$$\overline{\mathbf{K}}_{EB4DOF} = \begin{pmatrix} \frac{12EI}{L^3} & \frac{6EI}{L^2} & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{4EI}{L} & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{2EI}{L} & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{pmatrix} \quad (35)$$

4 DOF Euler-Bernoulli beam element in a nutshell

- Displacement in vertical direction (beam's deflection) is approximated using third order interpolation for v_h .
- Two nodes.
- Two DOF's (v, θ) at node.
- Third order interpolation

$$v_h = N_1 v_1 + N_2 \theta_1 + N_3 v_2 + N_4 \theta_2 = \mathbf{N} \mathbf{u}.$$

$$\mathbf{N} = \begin{pmatrix} N_1 & N_2 & N_3 & N_4 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} v_1 & \theta_1 & v_2 & \theta_2 \end{pmatrix}^T$$

$$N_1 = \frac{2x^3}{L^3} - \frac{3x^2}{L^2} + 1 \quad ; N_2 = \frac{x^3}{L^2} - \frac{2x^2}{L} + x \quad ; N_3 = -\frac{2x^3}{L^3} + \frac{3x^2}{L^2} \quad ; N_4 = \frac{x^3}{L^2} - \frac{x^2}{L}$$

$$\bar{\mathbf{K}} = \begin{pmatrix} \frac{12EI}{L^3} & \frac{6EI}{L^2} & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{4EI}{L} & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{2EI}{L} & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{pmatrix}$$

6 DOF's Euler-Bernoulli beam element

- Displacement in axial and vertical direction are approximated using linear interpolation for u_h and third order interpolation for v_h .
- Two nodes, three DOF's (u, v, θ) at node.

$$\mathbf{u}_h = \begin{pmatrix} u_h \\ v_h \end{pmatrix} = \begin{pmatrix} N_1^{lin}u_1 + N_2^{lin}u_2 \\ N_1v_1 + N_2\theta_1 + N_3v_2 + N_4\theta_2 \end{pmatrix} \quad (36)$$

$$\mathbf{N} = \begin{pmatrix} N_1^{lin} & 0 & 0 & N_1^{lin} & 0 & 0 \\ 0 & N_1 & N_2 & 0 & N_3 & N_4 \end{pmatrix} \quad \mathbf{u} = (u_1 \quad v_1 \quad \theta_1 \quad u_2 \quad v_2 \quad \theta_2)^T$$

where

$$N_1^{lin} = 1 - x/L \quad ; \quad N_2^{lin} = x/L \quad (37)$$

and

$$N_1 = \frac{2x^3}{L^3} - \frac{3x^2}{L^2} + 1 \quad ; \quad N_2 = \frac{x^3}{L^2} - \frac{2x^2}{L} + x \quad ; \quad N_3 = -\frac{2x^3}{L^3} + \frac{3x^2}{L^2} \quad ; \quad N_4 = \frac{x^3}{L^2} - \frac{x^2}{L}$$

6 DOF's Euler-Bernoulli beam element: Energy

Strain energy due to bending deformation:

$$W_{int}^{bend} = \frac{1}{2} \int_V \sigma \varepsilon dV = \frac{1}{2} E \int_0^L \varepsilon_{xx}^2 \int_A y^2 dA dx = \frac{1}{2} EI \int_0^L \varepsilon_{xx}^2 dx \quad (38)$$

where I is moment of inertia and ε_{xx}^1 is curvature.

$$\varepsilon_{xx}^1 \approx \frac{d\theta}{dx} = \frac{d^2 v_h}{dx^2} = \frac{\partial^2 \mathbf{N}^b}{\partial x^2} \mathbf{u}_b \quad (39)$$

$$W_{int}^{axial} = \frac{1}{2} EA \int_x \varepsilon_{xx}^2 dx = \frac{1}{2} \bar{\mathbf{u}}_a^T \underbrace{EA \int_0^L \frac{\partial \mathbf{N}^{lin T}}{\partial x} \frac{\partial \mathbf{N}^{lin}}{\partial x} dx}_{\bar{\mathbf{K}}_{rod}} \bar{\mathbf{u}}_a \quad (40)$$

$$W_{int}^{bend} = \frac{1}{2} \bar{\mathbf{u}}_b^T \underbrace{EI \int_0^L \frac{\partial^2 \mathbf{N}^{b T}}{\partial x^2} \frac{\partial^2 \mathbf{N}^b}{\partial x^2} dx}_{\bar{\mathbf{K}}_{EB4DOF}} \bar{\mathbf{u}}_b \quad (41)$$

6 DOF's Euler-Bernoulli beam element: Elemental stiffness matrix

$$\overline{\mathbf{K}} = \begin{pmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{pmatrix} \quad (42)$$

6 DOF's Euler-Bernoulli beam element: Transformation

$$\mathbf{T} = \begin{pmatrix} \cos \alpha & \sin \alpha & 0 & 0 & 0 & 0 \\ -\sin \alpha & \cos \alpha & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos \alpha & \sin \alpha & 0 \\ 0 & 0 & 0 & -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (43)$$

$$\mathbf{K}_G = \mathbf{T}^T \overline{\mathbf{K}} \mathbf{T} \quad (44)$$

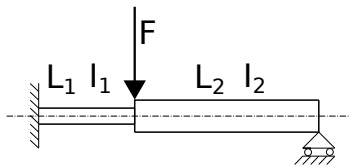
$$\overline{\mathbf{u}} = \mathbf{T} \mathbf{u} \quad (45)$$

$$\overline{\mathbf{u}} = \mathbf{T} \mathbf{u}; \quad \mathbf{u} = \mathbf{T}^T \overline{\mathbf{u}} \quad (46)$$

$$\overline{\mathbf{f}} = \mathbf{T} \mathbf{f}; \quad \mathbf{f} = \mathbf{T}^T \overline{\mathbf{f}} \quad (47)$$

Example: Beam structure with two beam elements

The aim is to solve nodal displacements and member forces using a two-dimensional **two-noded Euler-Bernoulli beam element (6 DOF)** in MATLAB. For simplicity, a mesh of two beam elements is only used to analyze the beam structure shown in the figure.



- Discretized by two 6 DOF's E-B elements
- 3 nodes, 3 DOF's at node, $3 \times 3 = 9$ DOF's
- 4 fixed/constrained nodal displacements.

Elemental stiffness matrices

Element 1:

$$\overline{\mathbf{K}}^{(1)} = \begin{matrix} & \begin{matrix} u_1 & v_1 & \theta_1 & u_2 & v_2 & \theta_2 \end{matrix} \\ \begin{pmatrix} \frac{E_1 A_1}{L_1} & 0 & 0 & -\frac{E_1 A_1}{L_1} & 0 & 0 \\ 0 & \frac{12 E_1 I_1}{L_1^3} & \frac{6 E_1 I_1}{L_1^2} & 0 & -\frac{12 E_1 I_1}{L_1^3} & \frac{6 E_1 I_1}{L_1^2} \\ 0 & \frac{6 E_1 I_1}{L_1^2} & \frac{4 E_1 I_1}{L_1} & 0 & -\frac{6 E_1 I_1}{L_1^2} & \frac{2 E_1 I_1}{L_1} \\ -\frac{E_1 A_1}{L_1} & 0 & 0 & \frac{E_1 A_1}{L_1} & 0 & 0 \\ 0 & -\frac{12 E_1 I_1}{L_1^3} & -\frac{6 E_1 I_1}{L_1^2} & 0 & \frac{12 E_1 I_1}{L_1^3} & -\frac{6 E_1 I_1}{L_1^2} \\ 0 & \frac{6 E_1 I_1}{L_1^2} & \frac{2 E_1 I_1}{L_1} & 0 & -\frac{6 E_1 I_1}{L_1^2} & \frac{4 E_1 I_1}{L_1} \end{pmatrix} & \begin{matrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{matrix} \end{matrix}$$

Elemental stiffness matrices

Element 2:

$$\overline{\mathbf{K}}^{(2)} = \begin{matrix} & \begin{matrix} u_2 & v_2 & \theta_2 & u_3 & v_3 & \theta_3 \end{matrix} \\ \begin{pmatrix} \frac{E_2 A_2}{L_2} & 0 & 0 & -\frac{E_2 A_2}{L_2} & 0 & 0 \\ 0 & \frac{12E_2 I_2}{L_2^3} & \frac{6E_2 I_2}{L_2^2} & 0 & -\frac{12E_2 I_2}{L_2^3} & \frac{6E_2 I_2}{L_2^2} \\ 0 & \frac{6E_2 I_2}{L_2^2} & \frac{4E_2 I_2}{L_2} & 0 & -\frac{6E_2 I_2}{L_2^2} & \frac{2E_2 I_2}{L_2} \\ -\frac{E_2 A_2}{L_2} & 0 & 0 & \frac{E_2 A_2}{L_2} & 0 & 0 \\ 0 & -\frac{12E_2 I_2}{L_2^3} & -\frac{6E_2 I_2}{L_2^2} & 0 & \frac{12E_2 I_2}{L_2^3} & -\frac{6E_2 I_2}{L_2^2} \\ 0 & \frac{6E_2 I_2}{L_2^2} & \frac{2E_2 I_2}{L_2} & 0 & -\frac{6E_2 I_2}{L_2^2} & \frac{4E_2 I_2}{L_2} \end{pmatrix} & \begin{matrix} u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \end{matrix} \end{matrix}$$

The transformation from a local coordinate system to a global coordinate system. Now $\theta = 0$ so the transformation is not necessarily needed because $\mathbf{T}$ is a unit matrix then.

$$\mathbf{K}_G^{(1)} = \mathbf{T}^T(0) \bar{\mathbf{K}}^{(1)} \mathbf{T}(0) \quad ; \quad \mathbf{K}_G^{(2)} = \mathbf{T}^T(0) \bar{\mathbf{K}}^{(2)} \mathbf{T}(0) \quad (48)$$

Assembling

$$K_G = \begin{pmatrix} u_1 & v_1 & \theta_1 & u_2 & v_2 & \theta_2 & u_3 & v_3 & \theta_3 \end{pmatrix} \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} & k_{15}^{(1)} & k_{16}^{(1)} & 0 & 0 & 0 \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} & k_{25}^{(1)} & k_{26}^{(1)} & 0 & 0 & 0 \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} & k_{35}^{(1)} & k_{36}^{(1)} & 0 & 0 & 0 \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} + k_{11}^{(2)} & k_{45}^{(1)} + k_{12}^{(2)} & k_{46}^{(1)} + k_{13}^{(2)} & k_{14}^{(2)} & k_{15}^{(2)} & k_{16}^{(2)} \\ k_{51}^{(1)} & k_{52}^{(1)} & k_{53}^{(1)} & k_{54}^{(1)} + k_{21}^{(2)} & k_{55}^{(1)} + k_{22}^{(2)} & k_{56}^{(1)} + k_{23}^{(2)} & k_{24}^{(2)} & k_{25}^{(2)} & k_{26}^{(2)} \\ k_{61}^{(1)} & k_{62}^{(1)} & k_{63}^{(1)} & k_{64}^{(1)} + k_{31}^{(2)} & k_{65}^{(1)} + k_{32}^{(2)} & k_{66}^{(1)} + k_{33}^{(2)} & k_{34}^{(2)} & k_{35}^{(2)} & k_{36}^{(2)} \\ 0 & 0 & 0 & k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} & k_{45}^{(2)} & k_{46}^{(2)} \\ 0 & 0 & 0 & k_{51}^{(2)} & k_{52}^{(2)} & k_{53}^{(2)} & k_{54}^{(2)} & k_{55}^{(2)} & k_{56}^{(2)} \\ 0 & 0 & 0 & k_{61}^{(2)} & k_{62}^{(2)} & k_{63}^{(2)} & k_{64}^{(2)} & k_{65}^{(2)} & k_{66}^{(2)} \end{pmatrix} \begin{pmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \end{pmatrix}$$

Force vector

$$\mathbf{f}_G = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -F \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{matrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \end{matrix}$$

Elimination of linear constraints

Fixed nodal displacements: $u_1 = v_1 = \theta_1 = v_3 = 0$. Let's eliminate corresponding rows and columns regarding FE-matrices.

$$\mathbf{f}_{G_{red}} = \begin{pmatrix} 0 \\ -F \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{matrix} u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ \theta_3 \end{matrix}$$

$$\mathbf{K}_{G_{red}} = \begin{pmatrix}
 \begin{matrix} u_2 \\ v_2 \\ \theta_2 \end{matrix} &
 \begin{matrix} u_2 \\ v_2 \\ \theta_2 \end{matrix} &
 \begin{matrix} \theta_2 \\ u_3 \\ \theta_3 \end{matrix} &
 \begin{matrix} u_3 \\ \theta_3 \end{matrix} \\
 \begin{matrix} k_{44}^{(1)} + k_{11}^{(2)} \\ k_{54}^{(1)} + k_{21}^{(2)} \\ k_{64}^{(1)} + k_{31}^{(2)} \\ k_{41}^{(2)} \\ k_{61}^{(2)} \end{matrix} &
 \begin{matrix} k_{45}^{(1)} + k_{12}^{(2)} \\ k_{55}^{(1)} + k_{22}^{(2)} \\ k_{65}^{(1)} + k_{32}^{(2)} \\ k_{42}^{(2)} \\ k_{62}^{(2)} \end{matrix} &
 \begin{matrix} k_{46}^{(1)} + k_{13}^{(2)} \\ k_{56}^{(1)} + k_{23}^{(2)} \\ k_{66}^{(1)} + k_{33}^{(2)} \\ k_{43}^{(2)} \\ k_{63}^{(2)} \end{matrix} &
 \begin{matrix} k_{14}^{(2)} \\ k_{24}^{(2)} \\ k_{34}^{(2)} \\ k_{44}^{(2)} \\ k_{64}^{(2)} \end{matrix} &
 \begin{matrix} k_{16}^{(2)} \\ k_{26}^{(2)} \\ k_{36}^{(2)} \\ k_{46}^{(2)} \\ k_{66}^{(2)} \end{matrix} \\
 u_2 & v_2 & \theta_2 & u_3 & \theta_3
 \end{pmatrix}$$

Note that $\mathbf{K}_{G_{red}}$ is symmetric i.e. $k_{ij} = k_{ji}$ (if not, there is a typo) and $\det \mathbf{K}_{G_{red}} > 0$ (if not, check BC's).

And then we solve displacements (in a global coordinate system).

$$\mathbf{u}_{G_{red}} = \mathbf{K}_{G_{red}}^{-1} \mathbf{f}_{G_{red}}$$